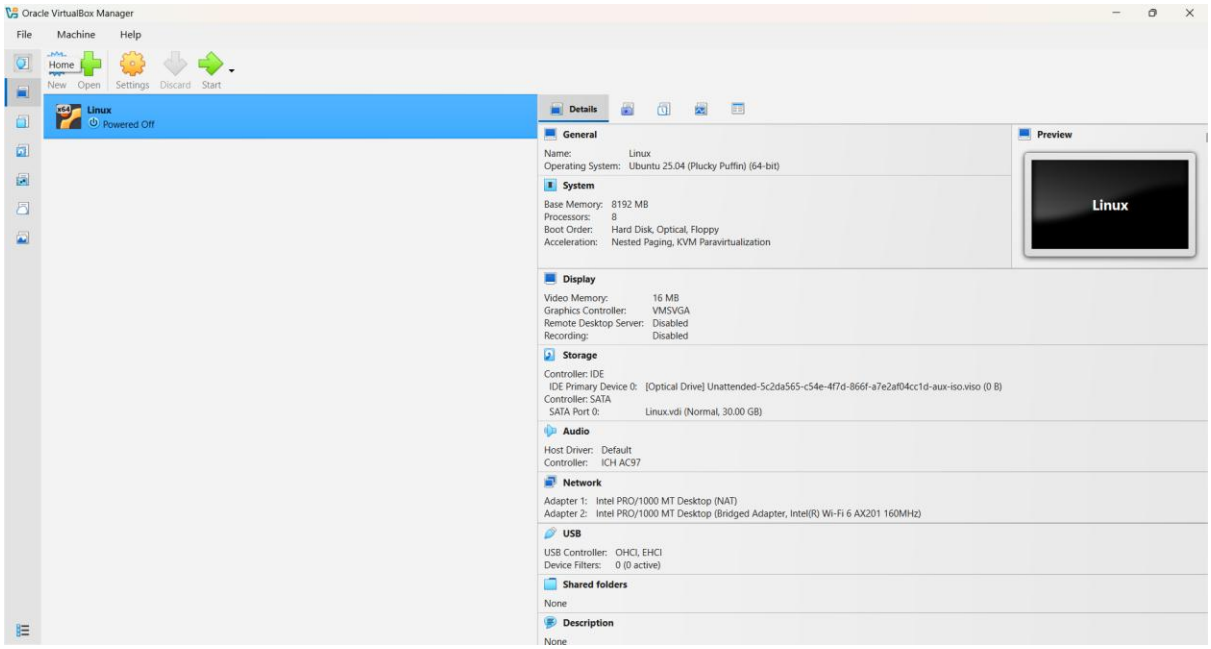
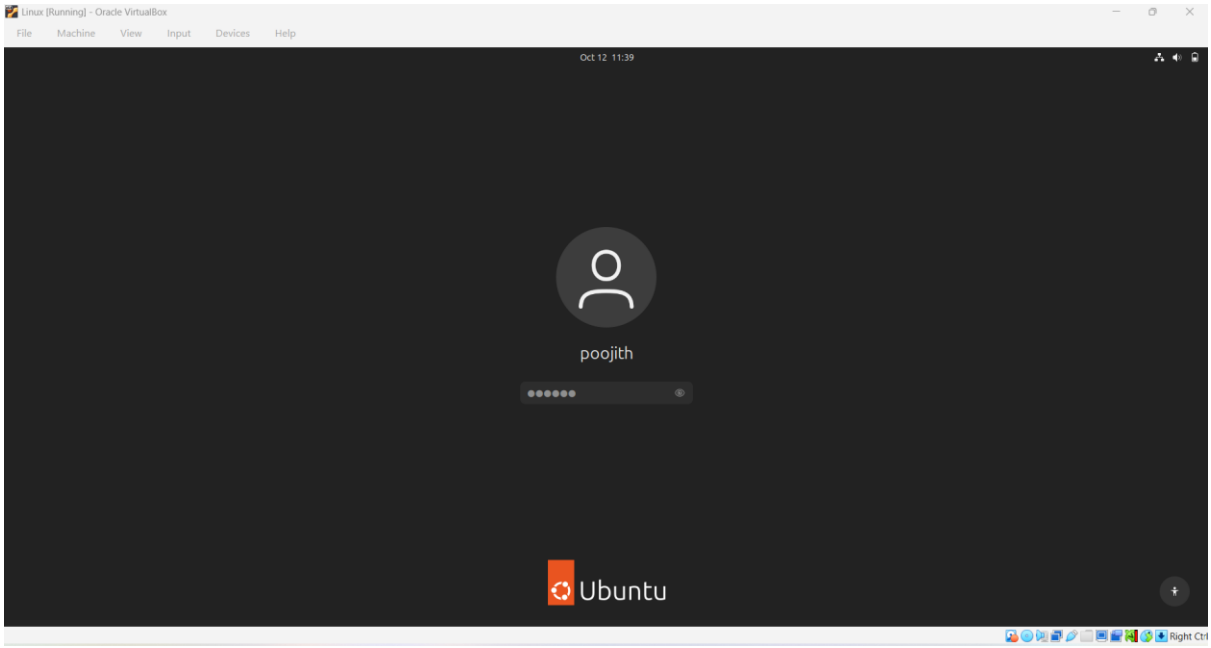
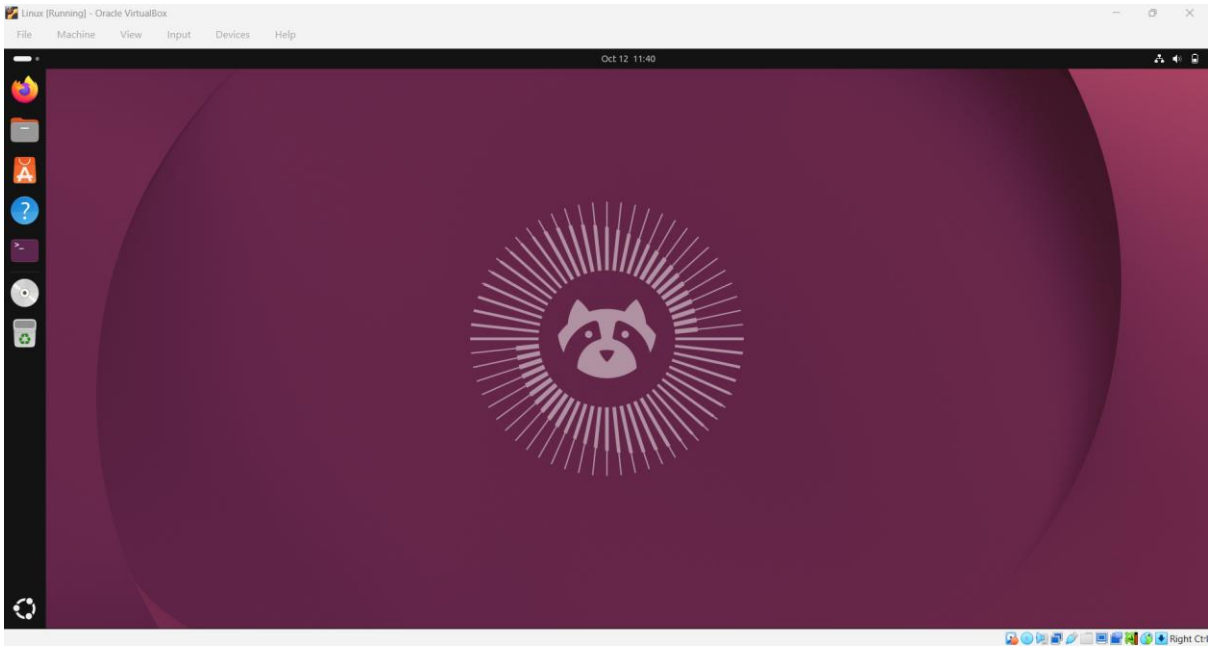
1. **Setup a virtual machine using Ubuntu image.**

Implementation Steps:

1. Install Oracle VM VirtualBox
2. Download the Ubuntu ISO image
3. Create a new virtual machine
    1. Enter the following details:
    2. Name: Ubuntu-VM
    3. Type: Linux
    4. Version: Ubuntu (64-bit)
    5. ISO Image: Select the downloaded Ubuntu ISO file.
    6. Configure the hardware resources. For example:
    7. RAM: 8 GB (8192 MB), if available.
    8. Processors: 4 CPUs, if available.
    9. Create a virtual hard disk:
    10. Disk size: 25 GB or more.
    11. Use the default virtual disk format unless your training requires otherwise.
    12. Click Finish.
4. Start the virtual machine
5. Install Ubuntu
6. Log in to Ubuntu







2. **Install docker on ec2.**

Implementation steps

1. Launch an EC2 instance using the Amazon Linux 2023 AMI.
2. Connect to the instance using EC2 Instance Connect or SSH.
3. Update the system packages.

sudo dnf update -y

4. Install Docker.

sudo dnf install -y docker

5. Start the Docker service.

sudo systemctl start docker

6. Enable Docker to start automatically when the instance reboots.

```
sudo systemctl enable docker
```

7. Verify the Docker service.

```
sudo systemctl status docker --no-pager
```

8. Check the Docker version.

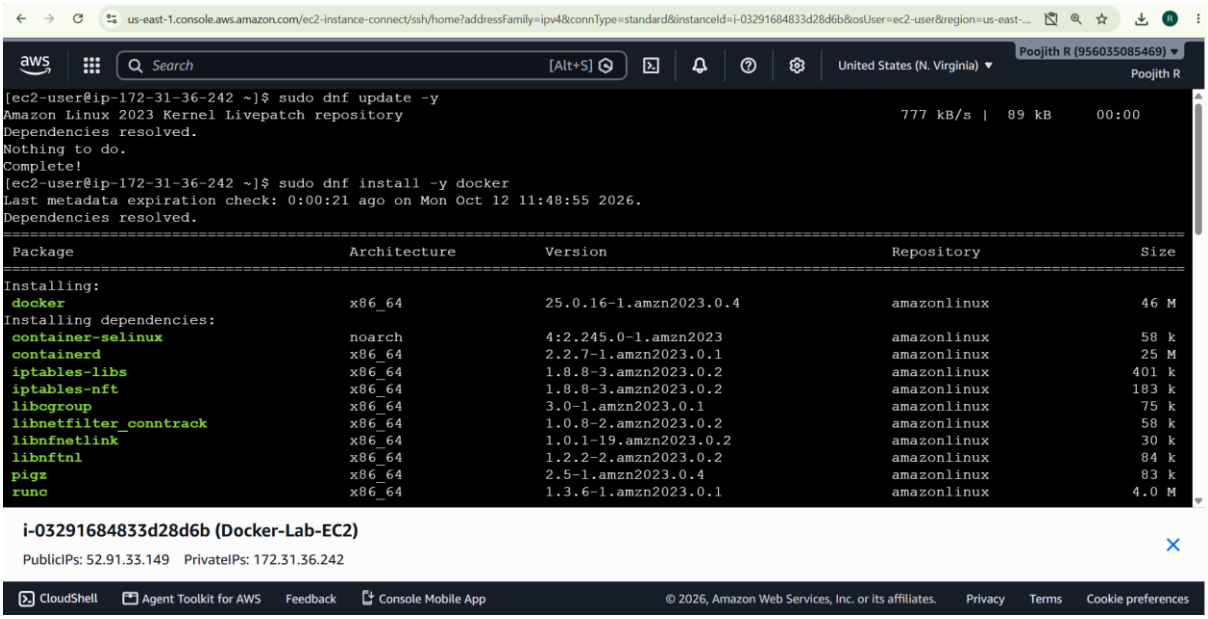
```
sudo docker --version
```

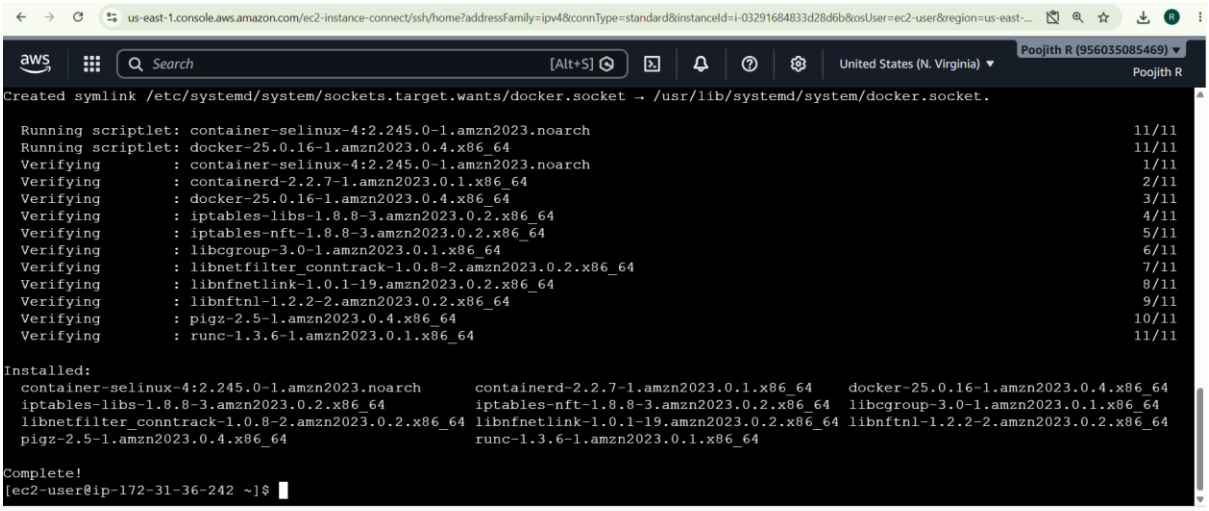
9. Run a test container.

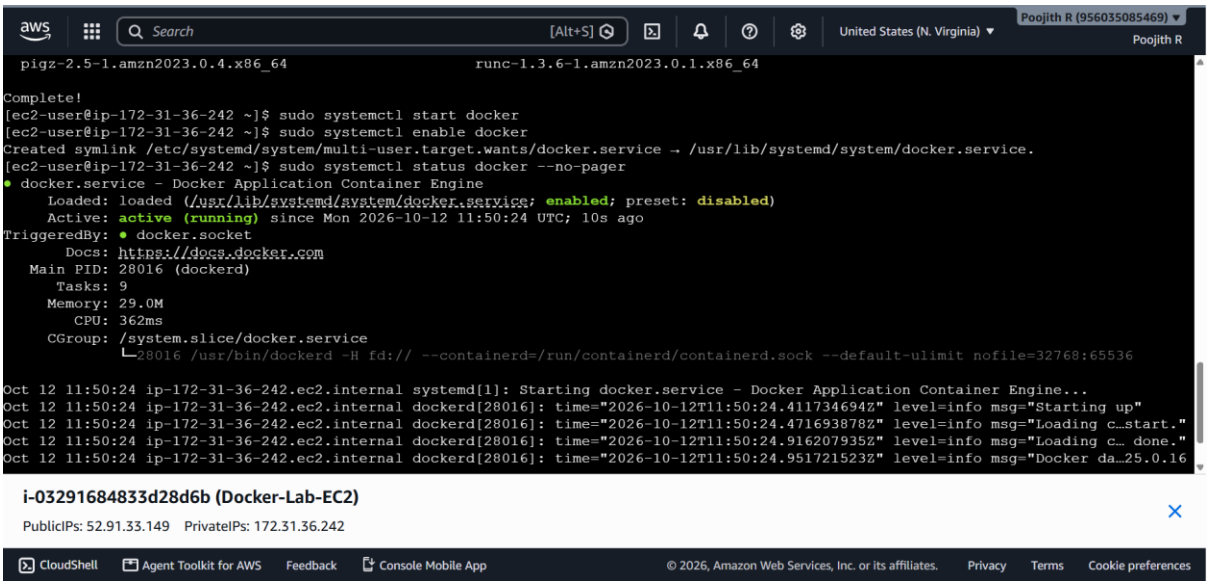
```
sudo docker run hello-world
```

Expected result

- Docker is installed successfully.
- The Docker service shows active (running).
- The test container displays Hello from Docker!.








```
[ec2-user@ip-172-31-36-242 ~]$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.
```

```
[ec2-user@ip-172-31-36-242 ~]$ sudo docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-36-242 ~]$
```

3. **Pull 5 docker images. (nginx, Apache tomcat, ubuntu, Jenkins, SonarQube)**

Implementation steps

Run the following commands one at a time.

1. Pull Nginx

sudo docker pull nginx:stable

2. Pull Apache Tomcat

sudo docker pull tomcat

3. Pull Ubuntu

sudo docker pull ubuntu:24.04

4. Pull Jenkins

Your original image was jenkins:2.60.3:

sudo docker pull jenkins:2.60.3

5. Pull SonarQube

sudo docker pull sonarqube:community

6. Verify the downloaded images

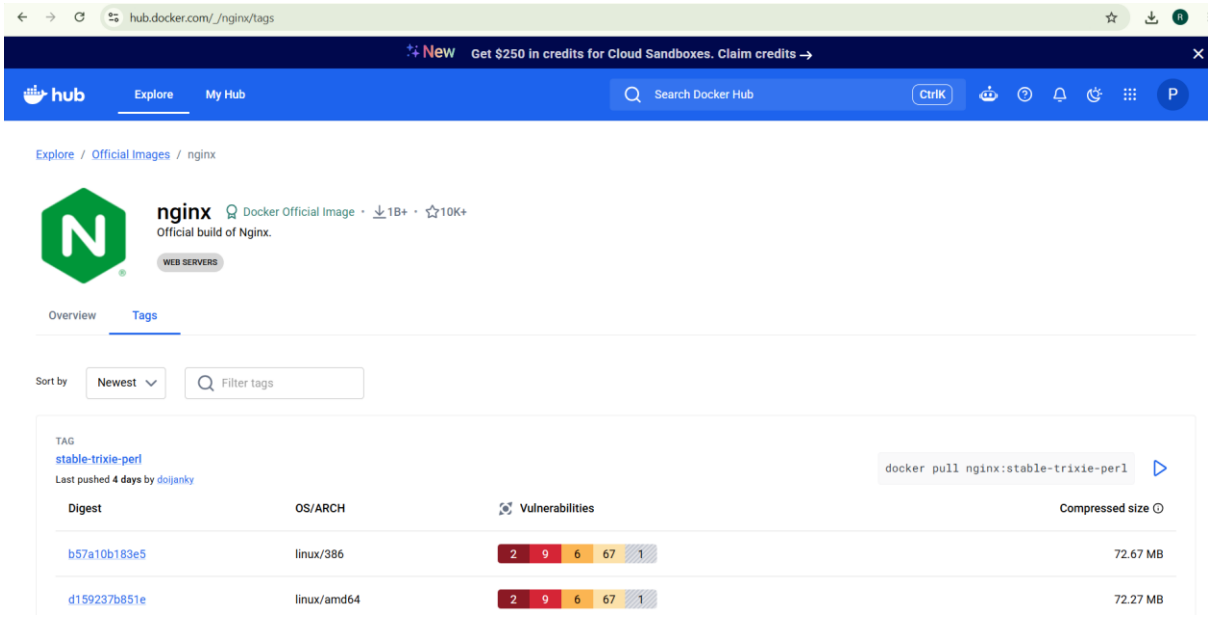
sudo docker images

Expected result

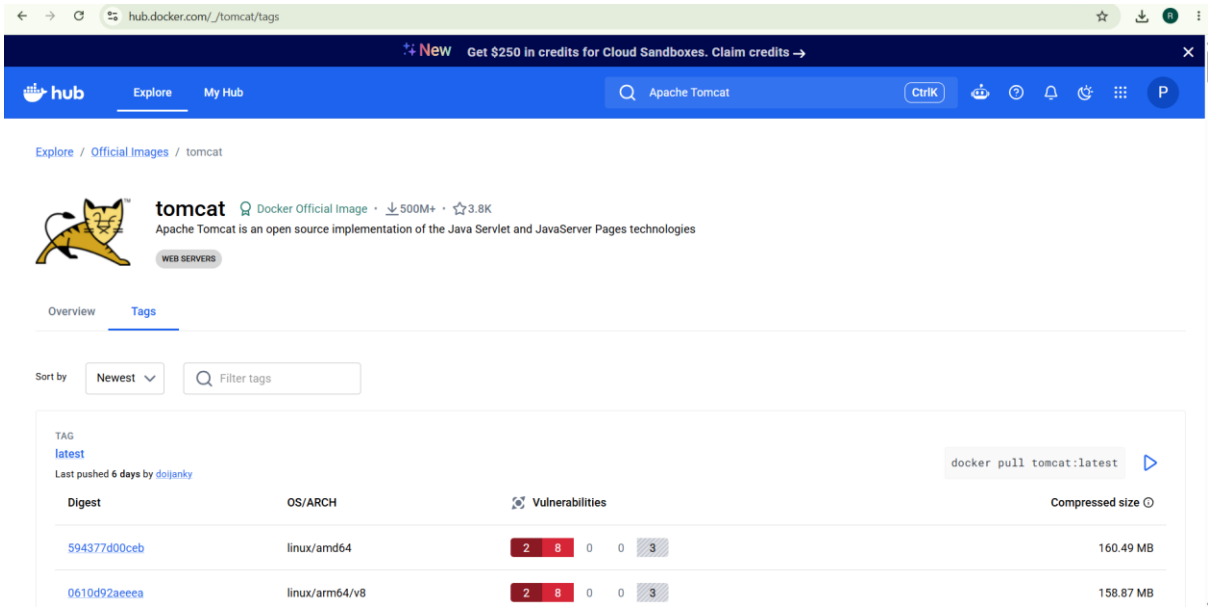
The image list contains Nginx, Tomcat, Ubuntu, Jenkins, and SonarQube. Your initial output showed these repositories and tags:

Repository	**Tag**
nginx	stable
tomcat	latest
ubuntu	24.04
jenkins	2.60.3
sonarqube	community

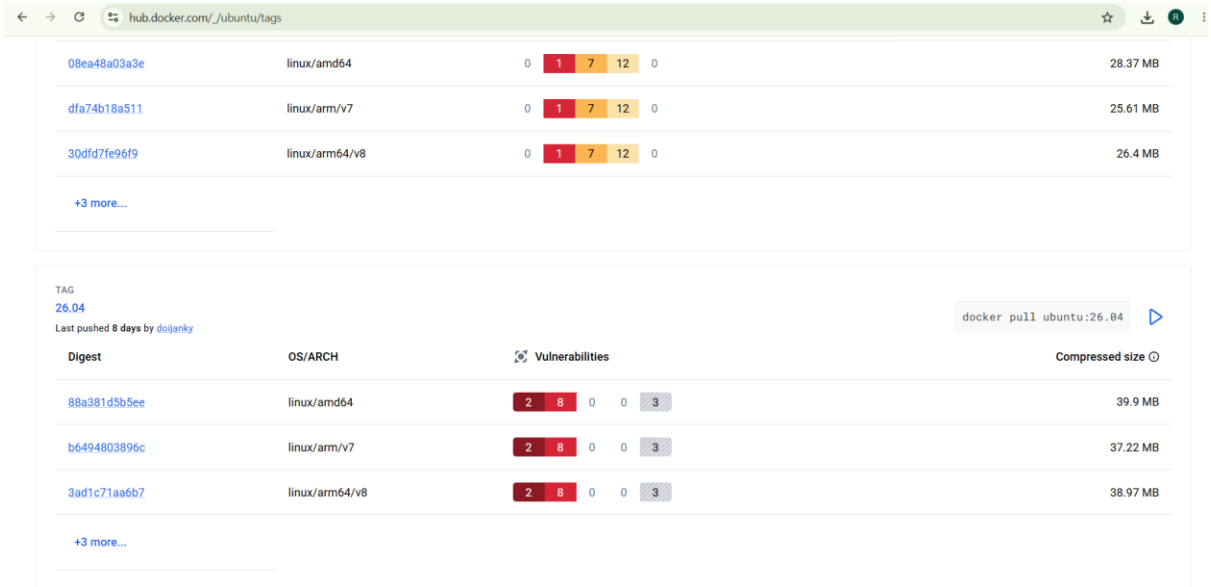




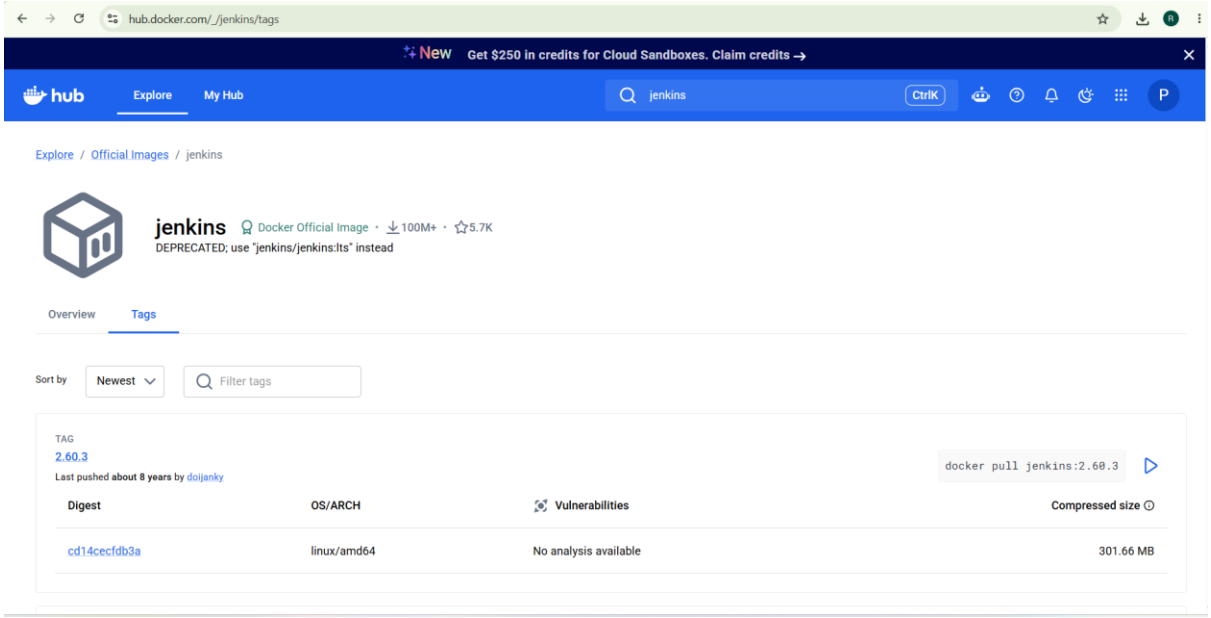

```
docker.io/library/nginx:stable
[ec2-user@ip-172-31-36-242 ~]$ sudo docker pull tomcat
Using default tag: latest
latest: Pulling from library/tomcat
4e07a0f12b2c: Pull complete
06ad70e463aa: Pull complete
5dec61325458: Pull complete
f39241fdc590: Pull complete
4f4fb700ef54: Pull complete
130739ca07b1: Pull complete
6d27111e20e8: Pull complete
a61c6f59ba30: Pull complete
Digest: sha256:bb7c0c78078841047e2d475acce833ef337cb0007ca7c7e4c9ff9058d9019801
Status: Downloaded newer image for tomcat:latest
```

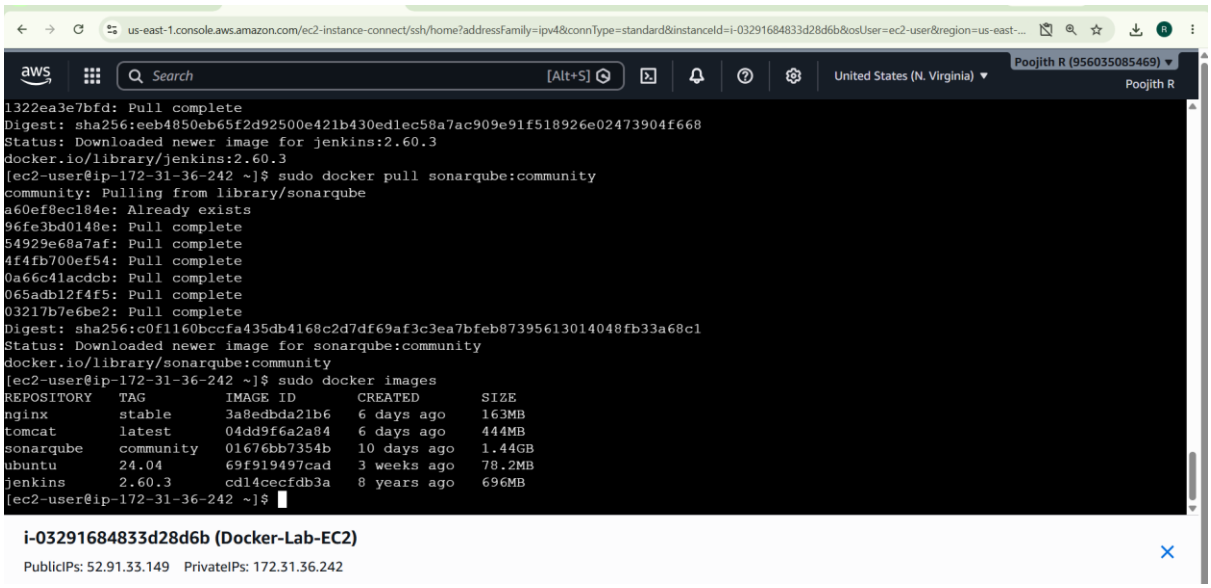


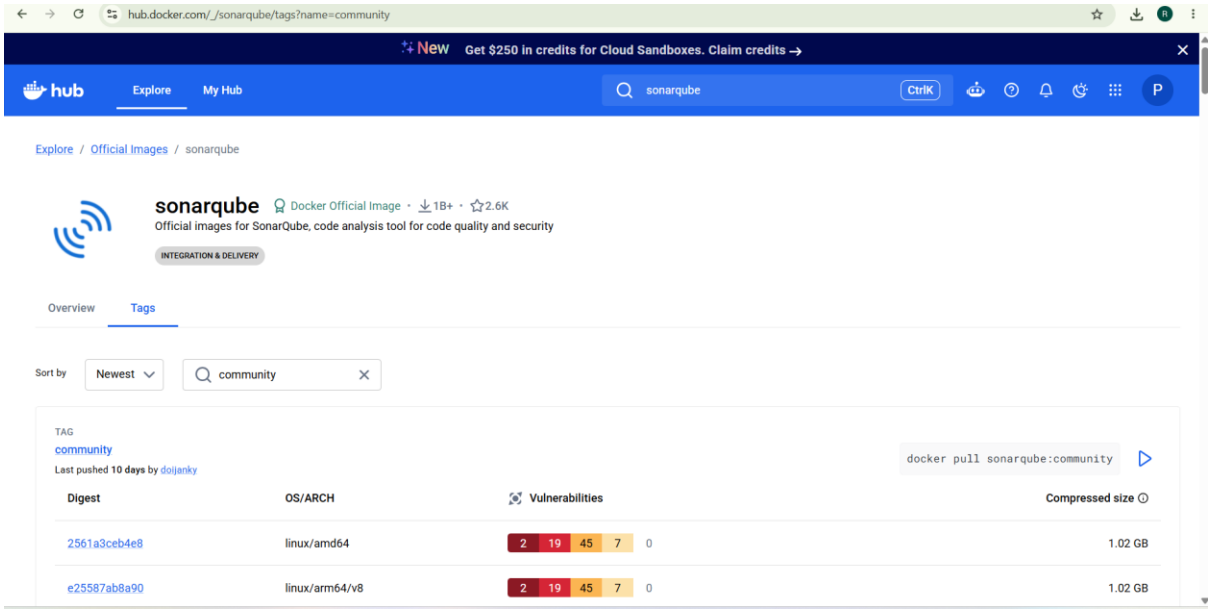

```
[ec2-user@ip-172-31-36-242 ~]$ sudo docker pull ubuntu:24.04
24.04: Pulling from library/ubuntu
a60ef8ec184e: Pull complete
Digest: sha256:534baea6a22c03a63003dbc8dbe78fe34bc0d7e595d9a9dc9834884ff530eb55
Status: Downloaded newer image for ubuntu:24.04
docker.io/library/ubuntu:24.04
```




aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
[ec2-user@ip-172-31-36-242 ~]$ sudo docker pull jenkins:2.60.3
2.60.3: Pulling from library/jenkins
55cbf04beb70: Pull complete
1607093a898c: Pull complete
9a8ea045c926: Pull complete
d4eee24d4dac: Pull complete
c58988e753d7: Pull complete
794a04897db9: Pull complete
70fcfa476f73: Pull complete
0539c80a02be: Pull complete
54fefc6dcf80: Pull complete
911bc90e47a8: Pull complete
38430d93efed: Pull complete
7e46ccda148a: Pull complete
c0cbcb5ac747: Pull complete
35ade7a86a8e: Pull complete
aa433a6a56b1: Pull complete
841c1dd38d62: Pull complete
b865dcb08714: Pull complete
5a3779030005: Pull complete
12b47c68955c: Pull complete
1322ea3e7bfd: Pull complete
Digest: sha256:eeb4850eb65f2d92500e421b430ed1ec58a7ac909e91f518926e02473904f668
Status: Downloaded newer image for jenkins:2.60.3


hub.docker.com/_/jenkins/tags
New
Get $250 in credits for Cloud Sandboxes. Claim credits →
hub
Explore
My Hub
jenkins
CtrlK
Explore / Official Images / jenkins
jenkins
Docker Official Image
100M+
5.7K
DEPRECATED; use "jenkins/jenkins:lts" instead
Overview
Tags
Sort by
Newest
Filter tags
TAG
2.60.3
Last pushed about 8 years by doijanky
docker pull jenkins:2.60.3
Digest
OS/ARCH
Vulnerabilities
Compressed size
cd14cecfdb3a
linux/amd64
No analysis available
301.66 MB


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
1322ea3e7bfd: Pull complete
Digest: sha256:eeb4850eb65f2d92500e421b430ed1ec58a7ac909e91f518926e02473904f668
Status: Downloaded newer image for jenkins:2.60.3
docker.io/library/jenkins:2.60.3
[ec2-user@ip-172-31-36-242 ~]$ sudo docker pull sonarqube:community
community: Pulling from library/sonarqube
a60ef8ec184e: Already exists
96fe3bd0148e: Pull complete
54929e68a7af: Pull complete
4f4fb700ef54: Pull complete
0a66c41acdcb: Pull complete
065adb12f4f5: Pull complete
03217b7e6be2: Pull complete
Digest: sha256:c0f1160bccfa435db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1
Status: Downloaded newer image for sonarqube:community
docker.io/library/sonarqube:community
[ec2-user@ip-172-31-36-242 ~]$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx stable 3a8edbda21b6 6 days ago 163MB
tomcat latest 04dd9f6a2a84 6 days ago 444MB
sonarqube community 01676bb7354b 10 days ago 1.44GB
ubuntu 24.04 69f919497cad 3 weeks ago 78.2MB
jenkins 2.60.3 cd14cecfdb3a 8 years ago 696MB
[ec2-user@ip-172-31-36-242 ~]$
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242



4. **Run nginx container and expose on port 81.**

Implementation steps

1. Create and start the Nginx container

sudo docker run -d --name nginx-container -p 81:80 nginx:stable

Explanation:

- -d runs the container in the background.
- --name nginx-container assigns a name to the container.
- -p 81:80 maps EC2 host port 81 to container port 80.
- nginx:stable specifies the image.

2. Verify the running container

sudo docker ps

The output should show nginx-container with a port mapping similar to:

0.0.0.0:81->80/tcp

3. Test Nginx from the EC2 terminal

curl -I http://localhost:81

Expected response includes:

HTTP/1.1 200 OK

Server: nginx

4. Configure the EC2 security group

Ensure an inbound rule allows TCP port 81 from your IP address.

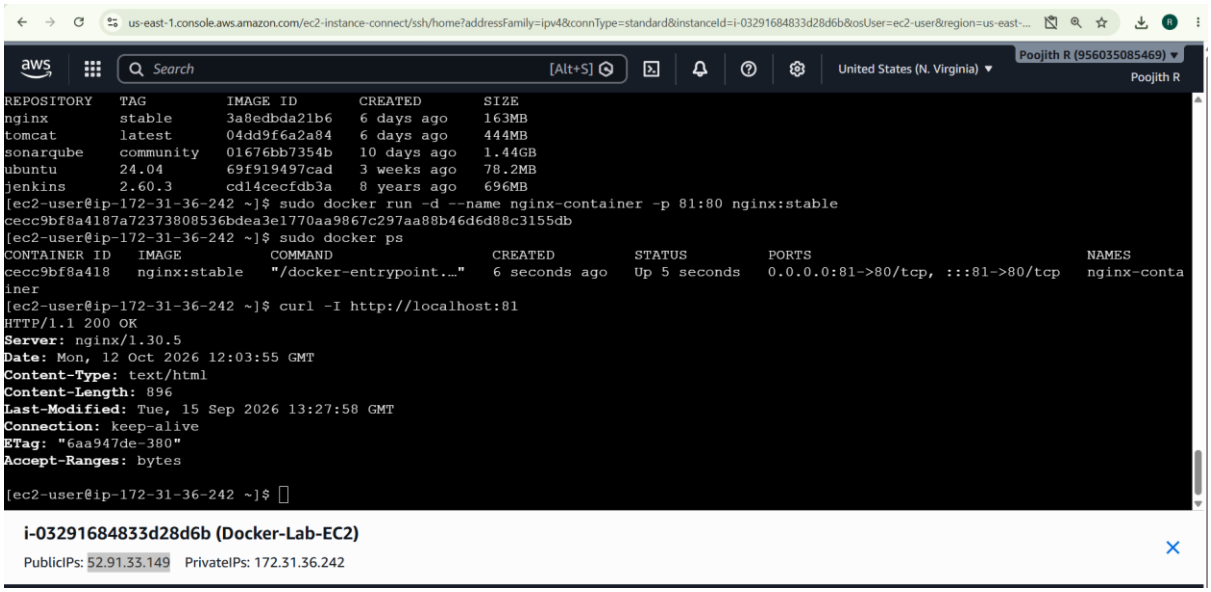
5. Open Nginx in a browser

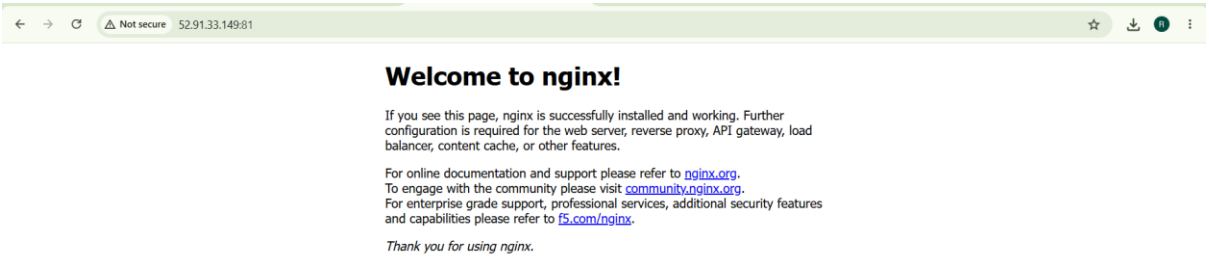
Use the following URL, replacing the placeholder with your EC2 public IPv4 address:

http://YOUR-EC2-PUBLIC-IP:81

Expected result

The browser displays the Welcome to nginx! page.





5. **Delete the Apache tomcat image from local.**

Implementation steps

1. Verify that the Tomcat image exists

sudo docker images tomcat

Your image used the latest tag.

2. Remove the image

sudo docker rmi tomcat:latest

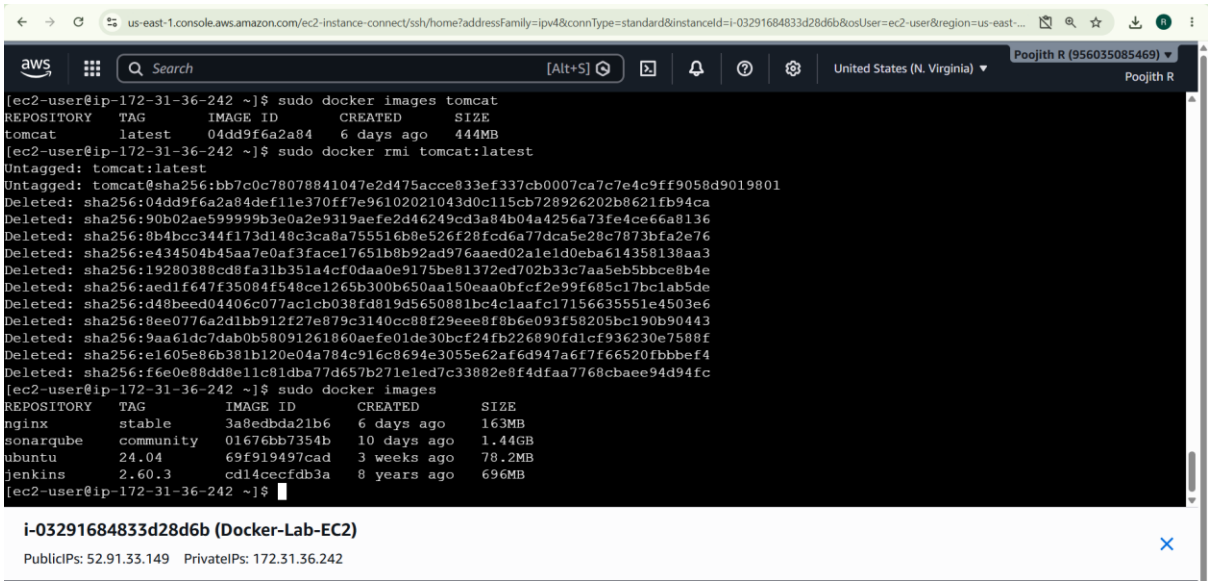
3. Verify the image list

sudo docker images

Confirm that the tomcat repository no longer appears.

Expected result

The local Tomcat image is removed successfully.



6. Inspect the Jenkins image, SonarQube image.

A. Inspect the Jenkins image

Your original Jenkins image was jenkins:2.60.3.

1. Display the complete image configuration

sudo docker image inspect jenkins:2.60.3

2. Display the image ID

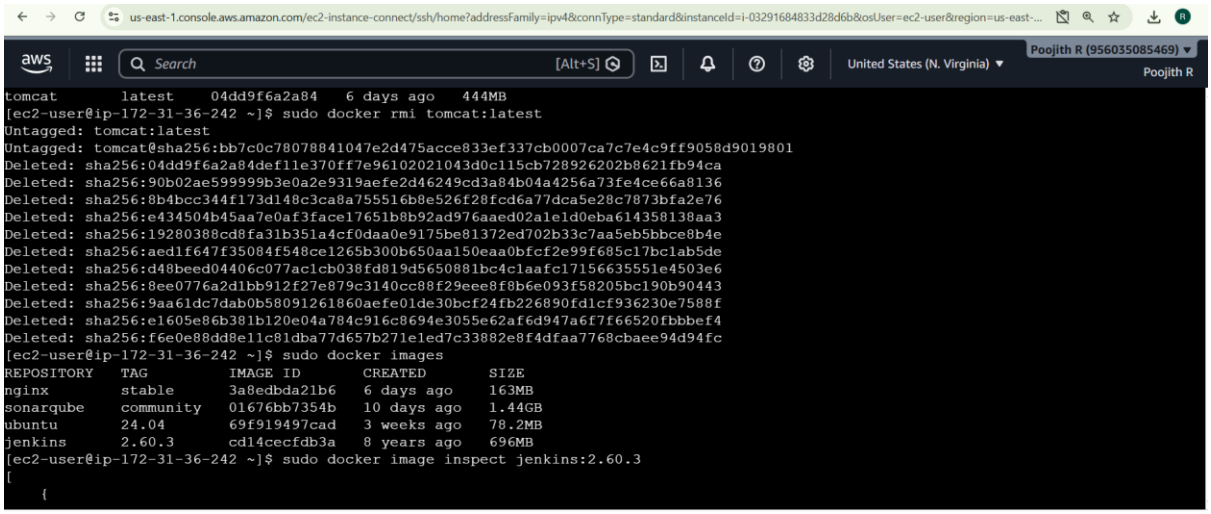
sudo docker image inspect jenkins:2.60.3 --format '{{.Id}}'

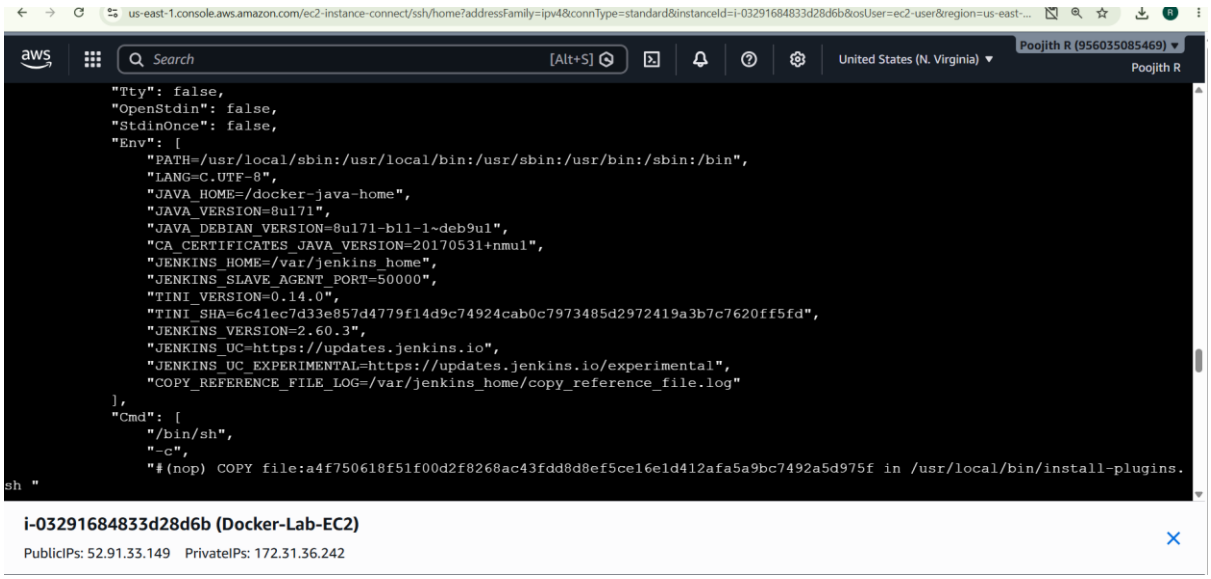
3. Display OS and architecture

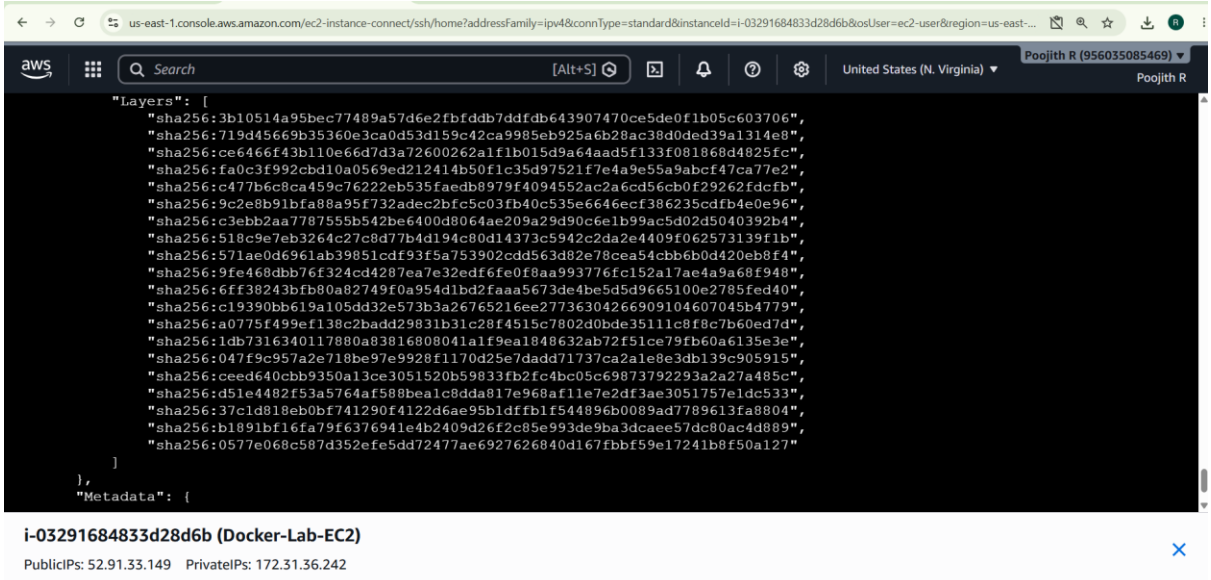
sudo docker image inspect jenkins:2.60.3 --format 'OS={{.Os}} Architecture={{.Architecture}}'

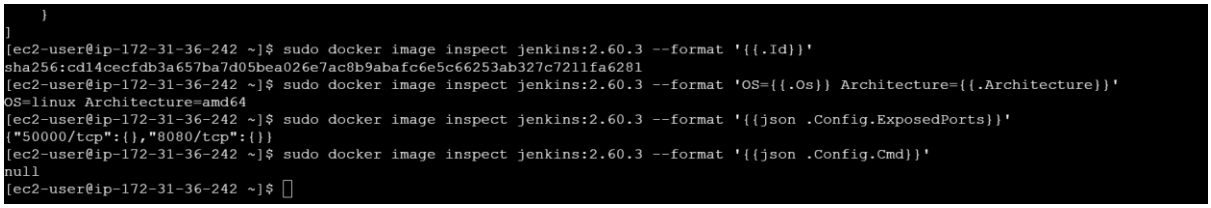
4. Display exposed ports

sudo docker image inspect jenkins:2.60.3 --format '{{json .Config.ExposedPorts}}'


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
tomcat latest 04dd9f6a2a84 6 days ago 444MB
[ec2-user@ip-172-31-36-242 ~]$ sudo docker rmi tomcat:latest
Untagged: tomcat:latest
Untagged: tomcat@sha256:bb7c0c78078841047e2d475acce833ef337cb0007ca7c7e4c9ff9058d9019801
Deleted: sha256:04dd9f6a2a84def11e370ff7e96102021043d0c115cb728926202b8621fb94ca
Deleted: sha256:90b02ae599999b3e0a2e9319aefe2d46249cd3a84b04a4256a73fe4ce66a8136
Deleted: sha256:8b4bcc344f173d148c3ca8a755516b8e526f28fcd6a77dca5e28c7873bfa2e76
Deleted: sha256:e434504b45aa7e0af3face17651b8b92ad976aaed02a1e1d0eba614358138aa3
Deleted: sha256:19280388cd8fa31b351a4cf0daa0e9175be81372ed702b33c7aa5eb5bbce8b4e
Deleted: sha256:aed1f647f35084f548ce1265b300b650aa150eaa0bfcf2e99f685c17bc1ab5de
Deleted: sha256:d48beed04406c077ac1cb038fd819d5650881bc4c1aafc17156635551e4503e6
Deleted: sha256:8ee0776a2d1bb912f27e879c3140cc88f29eee8f8b6e093f58205bc190b90443
Deleted: sha256:9aa61dc7dab0b58091261860aefe01de30bcf24fb226890fd1cf936230e7588f
Deleted: sha256:e1605e86b381b120e04a784c916c8694e3055e62af6d947a6f7f66520fbbbef4
Deleted: sha256:f6e0e88dd8e11c81dba77d657b271e1ed7c33882e8f4dfaa7768cbaee94d94fc
[ec2-user@ip-172-31-36-242 ~]$ sudo docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
nginx stable 3a8edbda21b6 6 days ago 163MB
sonarqube community 01676bb7354b 10 days ago 1.44GB
ubuntu 24.04 69f919497cad 3 weeks ago 78.2MB
jenkins 2.60.3 cd14cecfdb3a 8 years ago 696MB
[ec2-user@ip-172-31-36-242 ~]$ sudo docker image inspect jenkins:2.60.3
[
{


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
"Tty": false,
"OpenStdin": false,
"StdinOnce": false,
"Env": [
"PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
"LANG=C.UTF-8",
"JAVA_HOME=/docker-java-home",
"JAVA_VERSION=8u171",
"JAVA_DEBIAN_VERSION=8u171-b11-1~deb9u1",
"CA_CERTIFICATES_JAVA_VERSION=20170531+nmu1",
"JENKINS_HOME=/var/jenkins_home",
"JENKINS_SLAVE_AGENT_PORT=50000",
"TINI_VERSION=0.14.0",
"TINI_SHA=6c41ec7d33e857d4779f14d9c74924cab0c7973485d2972419a3b7c7620ff5fd",
"JENKINS_VERSION=2.60.3",
"JENKINS_UC=https://updates.jenkins.io",
"JENKINS_UC_EXPERIMENTAL=https://updates.jenkins.io/experimental",
"COPY_REFERENCE_FILE_LOG=/var/jenkins_home/copy_reference_file.log"
],
"Cmd": [
"/bin/sh",
"-c",
"#(nop) COPY file:a4f750618f51f00d2f8268ac43fdd8d8ef5ce16e1d412afa5a9bc7492a5d975f in /usr/local/bin/install-plugins.
sh "
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
"Layers": [
"sha256:3b10514a95bec77489a57d6e2fbfddb7ddfdb643907470ce5de0f1b05c603706",
"sha256:719d45669b35360e3ca0d53d159c42ca9985eb925a6b28ac38d0ded39a1314e8",
"sha256:ce6466f43b110e66d7d3a72600262a1f1b015d9a64aad5f133f081868d4825fc",
"sha256:fa0c3f992cbd10a0569ed212414b50f1c35d97521f7e4a9e55a9abcf47ca77e2",
"sha256:c477b6c8ca459c76222eb535faedb8979f4094552ac2a6cd56cb0f29262fdcfb",
"sha256:9c2e8b91bfa88a95f732adec2bfc5c03fb40c535e6646ecf386235cdfb4e0e96",
"sha256:c3ebb2aa7787555b542be6400d8064ae209a29d90c6e1b99ac5d02d5040392b4",
"sha256:518c9e7eb3264c27c8d77b4d194c80d14373c5942c2da2e4409f062573139f1b",
"sha256:571ae0d6961ab39851cdf93f5a753902cdd563d82e78cea54cbb6b0d420eb8f4",
"sha256:9fe468dbb76f324cd4287ea7e32edf6fe0f8aa993776fc152a17ae4a9a68f948",
"sha256:6ff38243bfb80a82749f0a954d1bd2faaa5673de4be5d5d9665100e2785fed40",
"sha256:c19390bb619a105dd32e573b3a26765216ee27736304266909104607045b4779",
"sha256:a0775f499ef138c2badd29831b31c28f4515c7802d0bde35111c8f8c7b60ed7d",
"sha256:1db731634011788 0a83816808041a1f9ea1848632ab72f51ce79fb60a6135e3e",
"sha256:047f9c957a2e718be97e9928f1170d25e7dadd71737ca2a1e8e3db139c905915",
"sha256:ceed640cbb9350a13ce3051520b59833fb2fc4bc05c69873792293a2a27a485c",
"sha256:d51e4482f53a5764af588bea1c8dda817e968af11e7e2df3ae3051757e1dc533",
"sha256:37c1d818eb0bf741290f4122d6ae95b1dffb1f544896b0089ad7789613fa8804",
"sha256:b1891bf16fa79f6376941e4b2409d26f2c85e993de9ba3dcaee57dc80ac4d889",
"sha256:0577e068c587d352efe5dd72477ae6927626840d167fbbf59e17241b8f50a127"
]
},
"Metadata": {
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242



B. Inspect the SonarQube image

1. Display the complete image configuration

sudo docker image inspect sonarqube:community

2. Display the image ID

sudo docker image inspect sonarqube:community --format '{{.Id}}'

3. Display OS and architecture

sudo docker image inspect sonarqube:community --format 'OS={{.Os}} Architecture={{.Architecture}}'

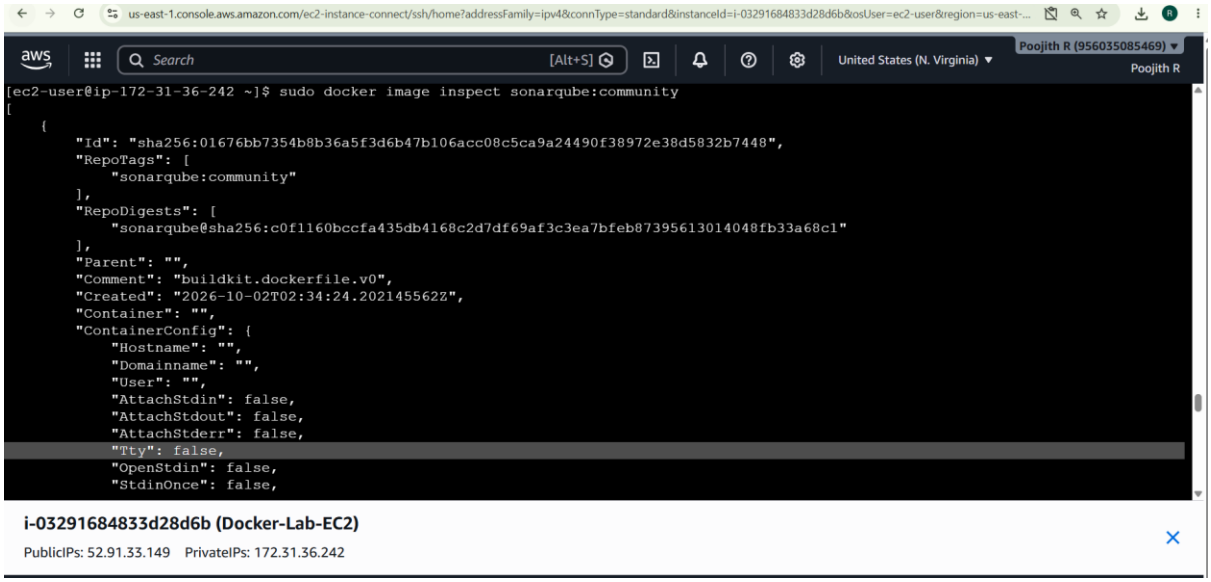
4. Display exposed ports

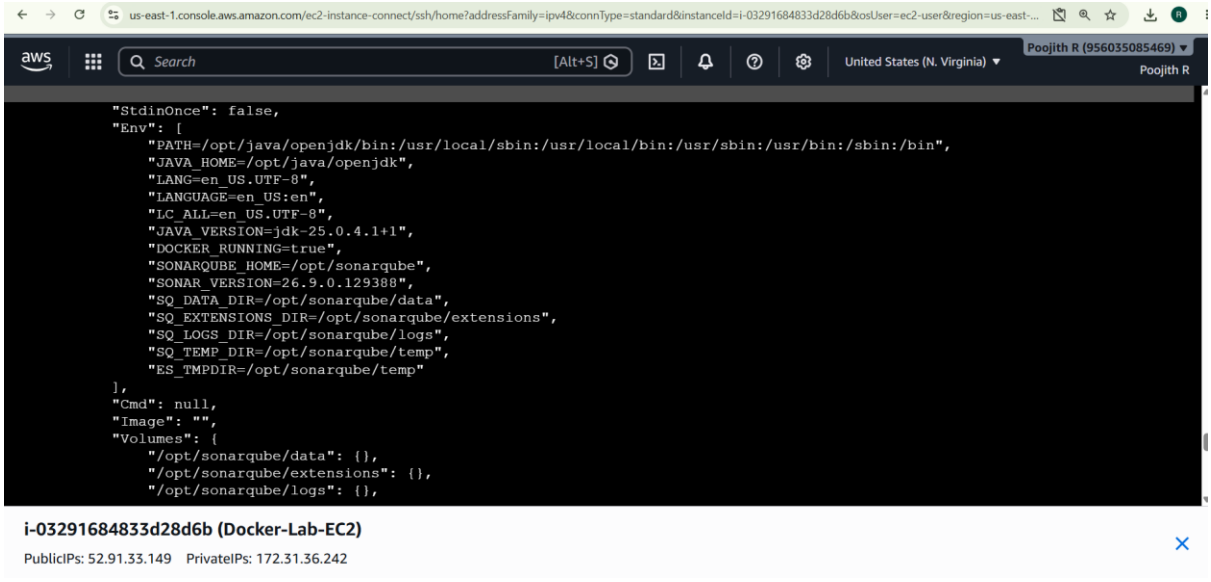
sudo docker image inspect sonarqube:community --format '{{json .Config.ExposedPorts}}'

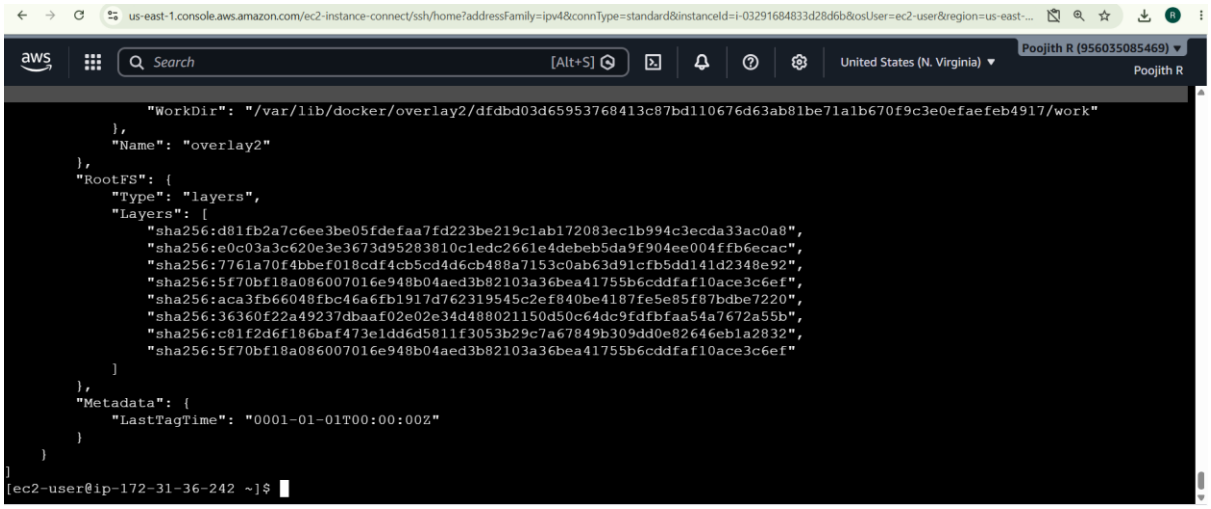
Expected result

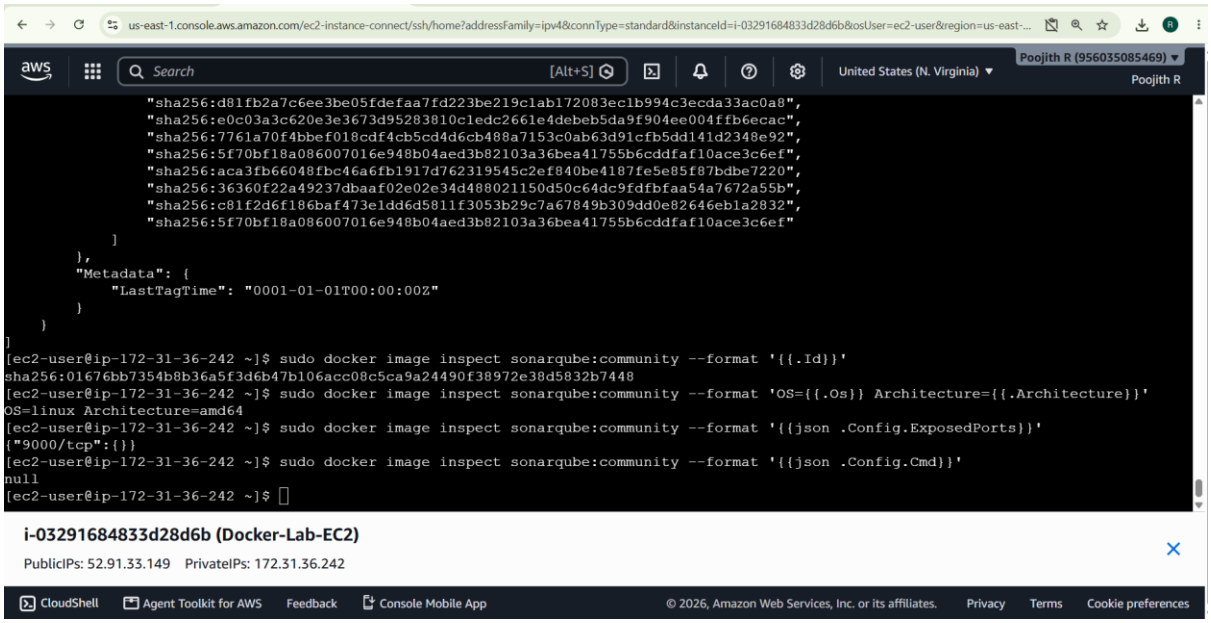
Docker displays metadata for both images, including their IDs, platform information, and configured ports.

Command or field	**Purpose**
docker image inspect	Displays detailed image metadata
.Id	Shows the image identifier
.Os	Shows the operating system recorded in the image
.Architecture	Shows the CPU architecture
.Config.ExposedPorts	Shows ports documented by the image


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
[ec2-user@ip-172-31-36-242 ~]$ sudo docker image inspect sonarqube:community
[
{
"Id": "sha256:01676bb7354b8b36a5f3d6b47b106acc08c5ca9a24490f38972e38d5832b7448",
"RepoTags": [
"sonarqube:community"
],
"RepoDigests": [
"sonarqube@sha256:c0f1160bccfa435db4168c2d7df69af3c3ea7bfeb87395613014048fb33a68c1"
],
"Parent": "",
"Comment": "buildkit.dockerfile.v0",
"Created": "2026-10-02T02:34:24.202145562Z",
"Container": "",
"ContainerConfig": {
"Hostname": "",
"Domainname": "",
"User": "",
"AttachStdin": false,
"AttachStdout": false,
"AttachStderr": false,
"Tty": false,
"OpenStdin": false,
"StdinOnce": false,
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
"StdinOnce": false,
"Env": [
"PATH=/opt/java/openjdk/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
"JAVA_HOME=/opt/java/openjdk",
"LANG=en_US.UTF-8",
"LANGUAGE=en_US:en",
"LC_ALL=en_US.UTF-8",
"JAVA_VERSION=jdk-25.0.4.1+1",
"DOCKER_RUNNING=true",
"SONARQUBE_HOME=/opt/sonarqube",
"SONAR_VERSION=26.9.0.129388",
"SQ_DATA_DIR=/opt/sonarqube/data",
"SQ_EXTENSIONS_DIR=/opt/sonarqube/extensions",
"SQ_LOGS_DIR=/opt/sonarqube/logs",
"SQ_TEMP_DIR=/opt/sonarqube/temp",
"ES_TMPDIR=/opt/sonarqube/temp"
],
"Cmd": null,
"Image": "",
"Volumes": {
"/opt/sonarqube/data": {},
"/opt/sonarqube/extensions": {},
"/opt/sonarqube/logs": {},
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242


us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
"WorkDir": "/var/lib/docker/overlay2/dfdbd03d65953768413c87bd110676d63ab81be71a1b670f9c3e0efaefeb4917/work"
},
"Name": "overlay2"
},
"RootFS": {
"Type": "layers",
"Layers": [
"sha256:d81fb2a7c6ee3be05fdefaa7fd223be219c1ab172083ec1b994c3ecda33ac0a8",
"sha256:e0c03a3c620e3e3673d95283810c1edc2661e4debeb5da9f904ee004ffb6ecac",
"sha256:7761a70f4bbef018cdf4cb5cd4d6cb488a7153c0ab63d91cfb5dd141d2348e92",
"sha256:5f70bf18a086007016e948b04aed3b82103a36bea41755b6cddfaf10ace3c6ef",
"sha256:aca3fb66048fbc46a6fb1917d762319545c2ef840be4187fe5e85f87bdbe7220",
"sha256:36360f22a49237dbaaf02e02e34d488021150d50c64dc9fdfbfaa54a7672a55b",
"sha256:c81f2d6f186baf473e1dd6d5811f3053b29c7a67849b309dd0e82646eb1a2832",
"sha256:5f70bf18a086007016e948b04aed3b82103a36bea41755b6cddfaf10ace3c6ef"
]
},
"Metadata": {
"LastTagTime": "0001-01-01T00:00:00Z"
}
}
]
[ec2-user@ip-172-31-36-242 ~]$



7. **Run Jenkins container and run one sample job.**

Implementation steps

1. Pull the Jenkins image

Download the maintained Jenkins image from Docker Hub.

sudo docker pull jenkins/jenkins:lts-jdk21

2. Create a Docker volume

Create persistent storage for Jenkins data.

sudo docker volume create jenkins_home

Verify the volume:

sudo docker volume ls

3. Run the Jenkins container

sudo docker run -d --name jenkins-container --restart unless-stopped -p 8080:8080 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts-jdk21

This command starts Jenkins in the background, maps port 8080 on EC2 to port 8080 inside the container, and stores Jenkins data in the persistent volume.

4. Verify the running container

sudo docker ps

Confirm that jenkins-container is running and port 8080 is mapped correctly.

5. Check Jenkins startup logs

sudo docker logs --tail 100 jenkins-container

Wait for Jenkins to finish starting.

6. Retrieve the initial administrator password

sudo docker exec jenkins-container cat /var/jenkins_home/secrets/initialAdminPassword

Copy the displayed password for the initial setup.

7. Access Jenkins through a web browser

- Open the AWS EC2 console and copy the instance's public IPv4 address.
- Ensure the EC2 security group allows inbound TCP port 8080 from your IP address.
- Open the following URL, replacing the placeholder with your actual public IP:

http://YOUR-EC2-PUBLIC-IP:8080

- Enter the initial administrator password.
- Click Install suggested plugins.
- Create an administrator account and complete the setup.

8. Create a Freestyle project

Under Build Steps, select Add build step → Execute shell.

Enter the following shell script:

echo "Docker Jenkins Sample Job"

echo "Build Number: $BUILD_NUMBER"

echo "Job Name: $JOB_NAME"

date

whoami

pwd

hostname

echo "Build completed successfully!"

7. Click Save.

9. Execute the sample job

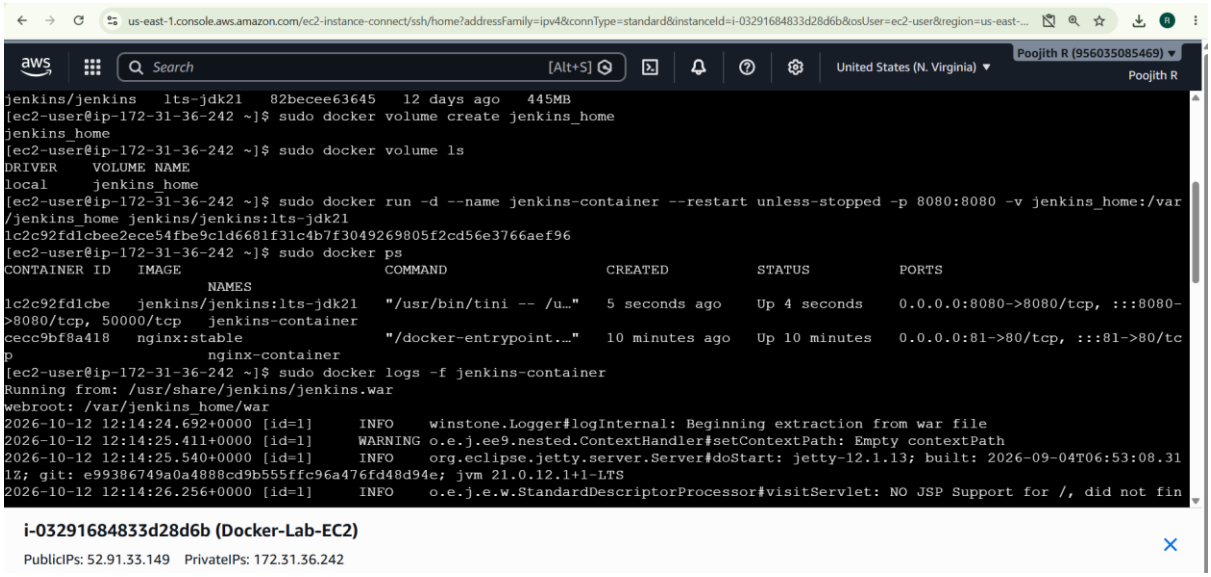
10.Click Console Output.

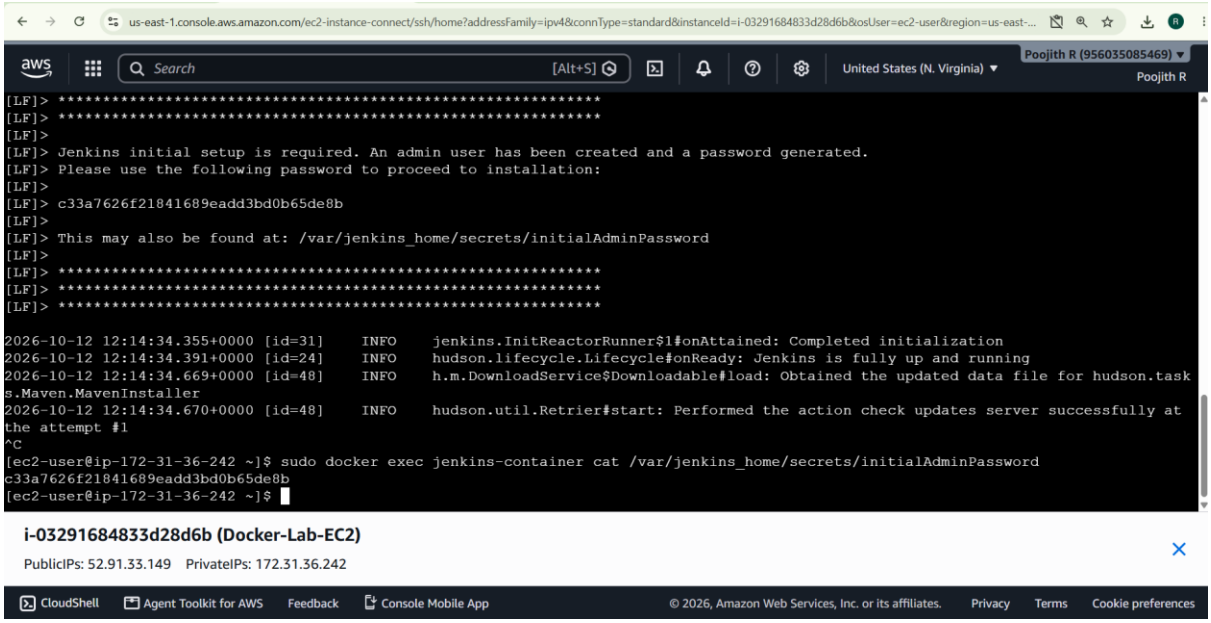
11. Verify that the output contains Build completed successfully! and the build status is successful.

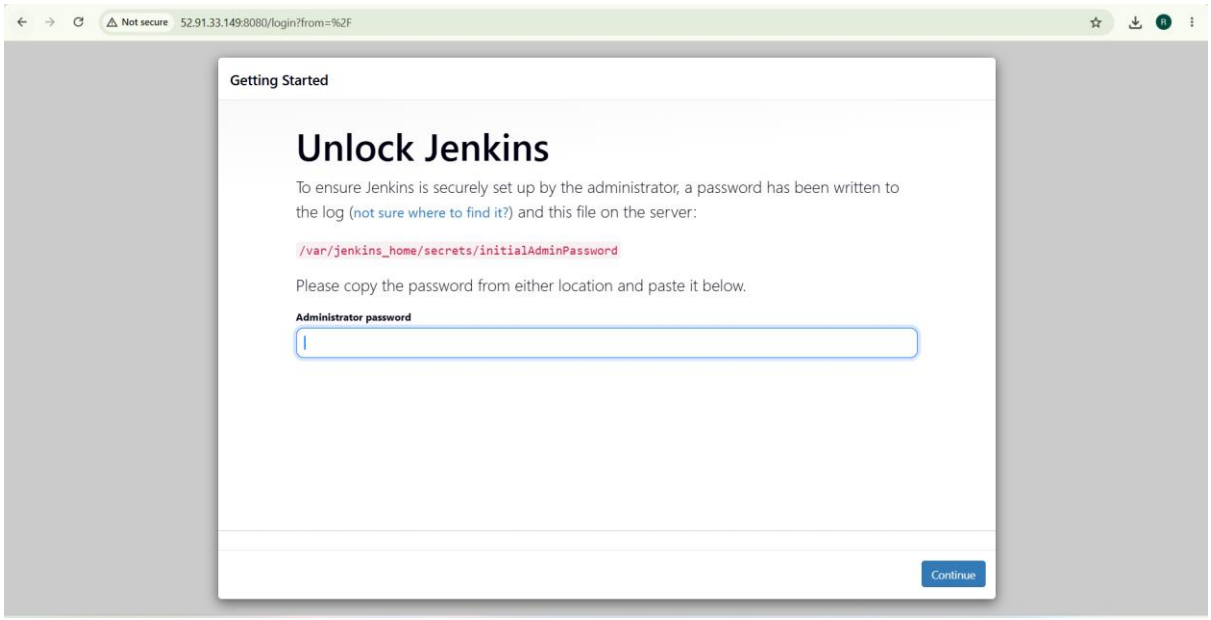
Expected result

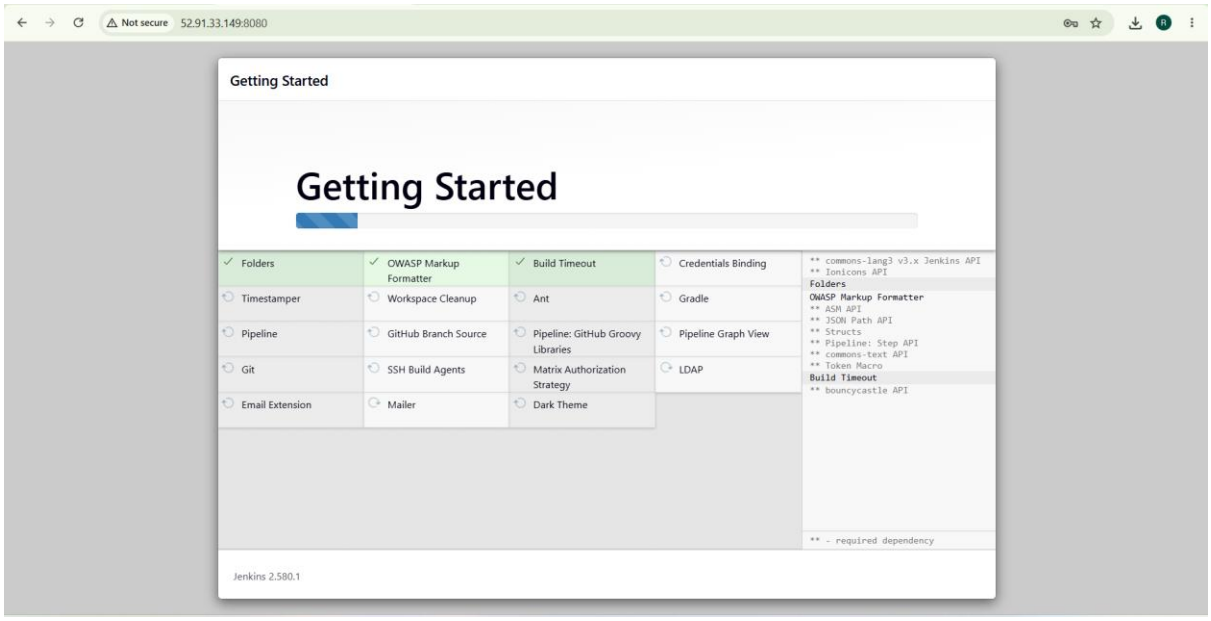
- Jenkins runs successfully inside a Docker container.
- The Jenkins web interface is accessible through port 8080.
- The Docker-Sample-Job Freestyle project executes successfully.
- The Console Output displays the build details and success message.

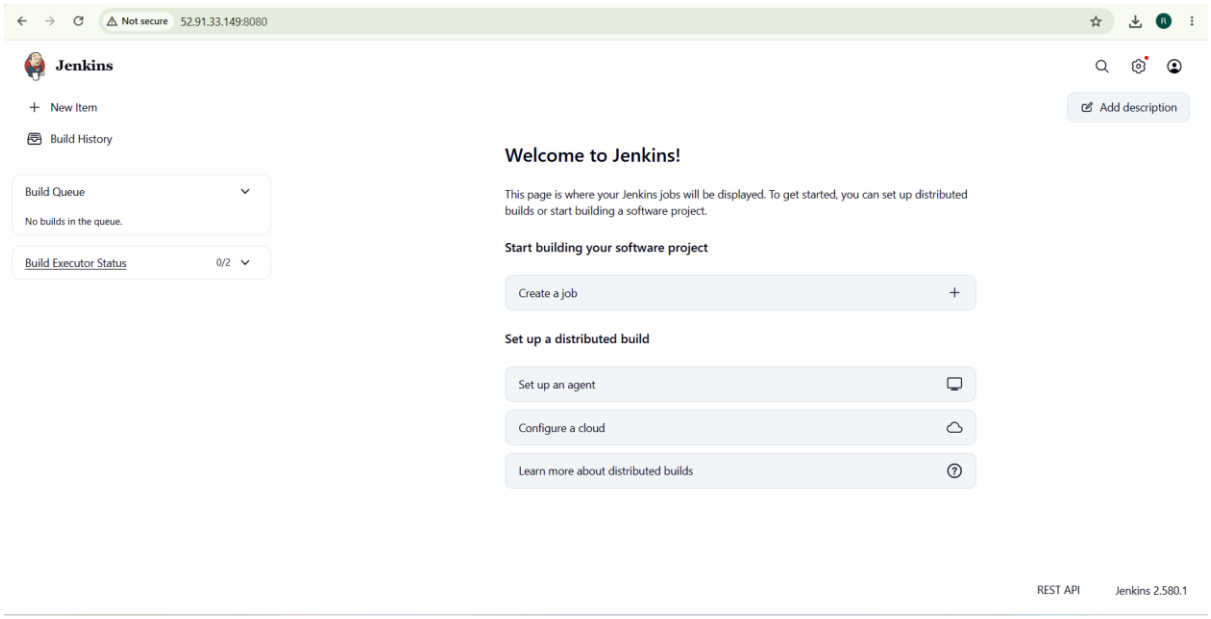


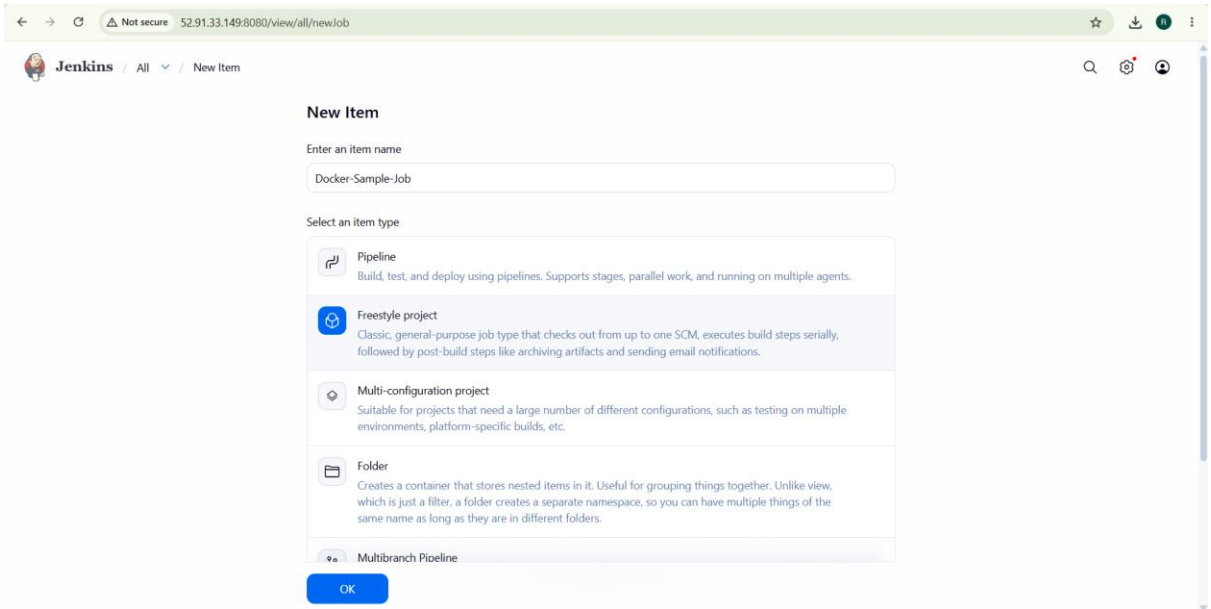


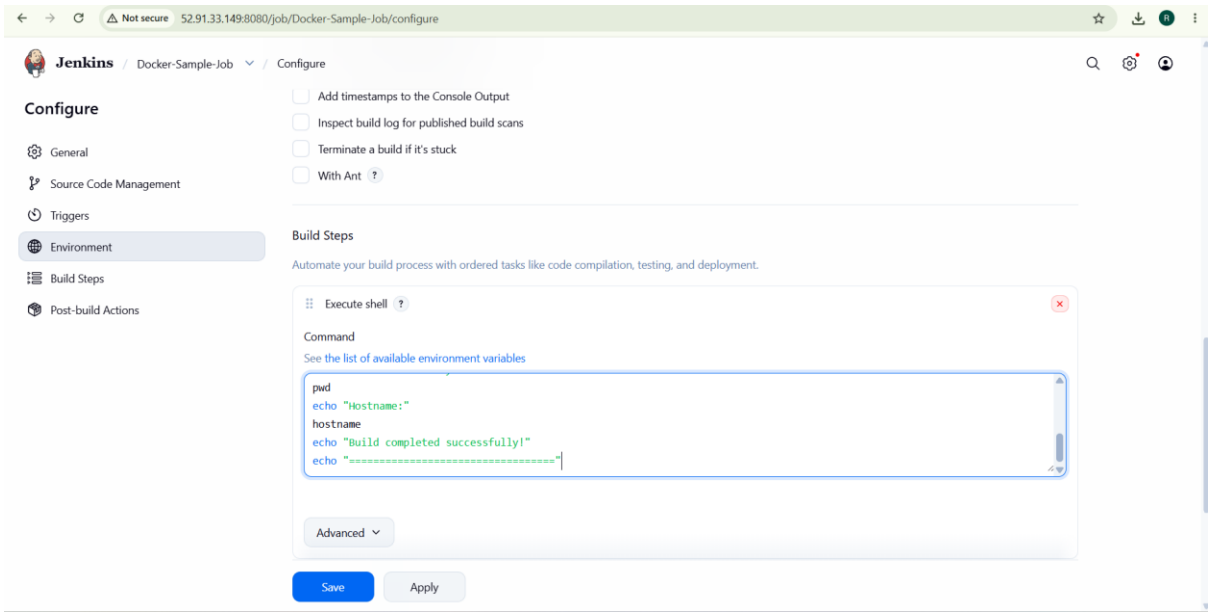

us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-03291684833d28d6b&osUser=ec2-user&region=us-east-...
aws
Search
[Alt+S]
United States (N. Virginia)
Poojith R (956035085469)
Poojith R
[LF]> **
[LF]> **
[LF]>
[LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> c33a7626f21841689eadd3bd0b65de8b
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> **
[LF]> **
[LF]> **
2026-10-12 12:14:34.355+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-10-12 12:14:34.391+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-10-12 12:14:34.669+0000 [id=48] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
2026-10-12 12:14:34.670+0000 [id=48] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
^C
[ec2-user@ip-172-31-36-242 ~]$ sudo docker exec jenkins-container cat /var/jenkins_home/secrets/initialAdminPassword
c33a7626f21841689eadd3bd0b65de8b
[ec2-user@ip-172-31-36-242 ~]$
i-03291684833d28d6b (Docker-Lab-EC2)
PublicIPs: 52.91.33.149 PrivateIPs: 172.31.36.242
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

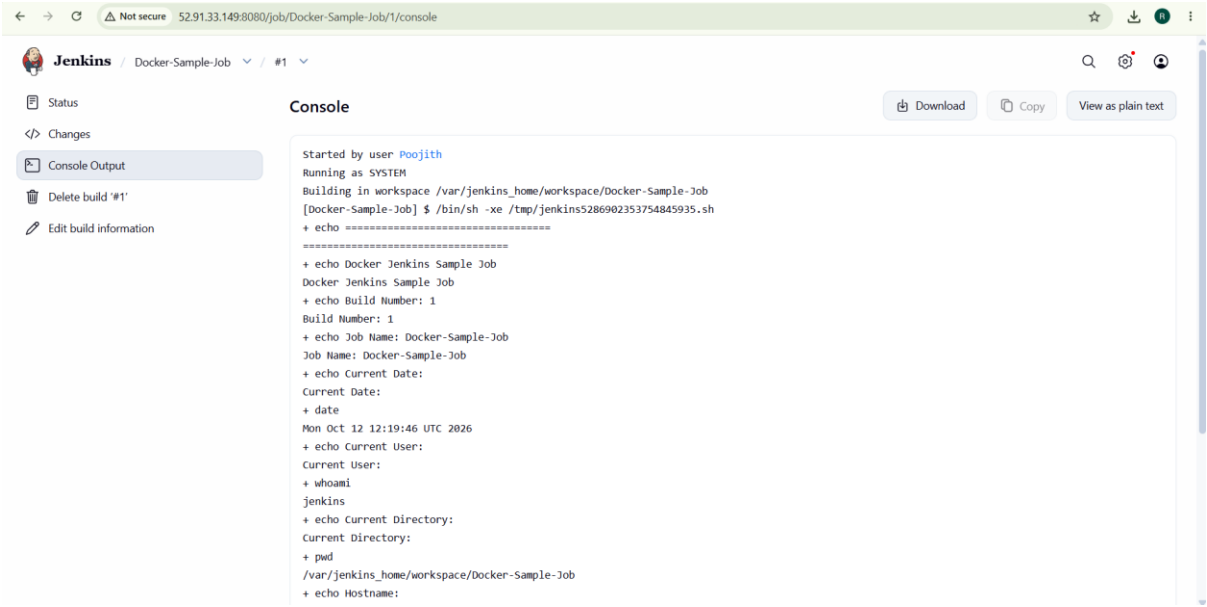

Not secure
52.91.33.149:8080/login?from=%2F
Getting Started
Unlock Jenkins
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:
/var/jenkins_home/secrets/initialAdminPassword
Please copy the password from either location and paste it below.
Administrator password
Continue

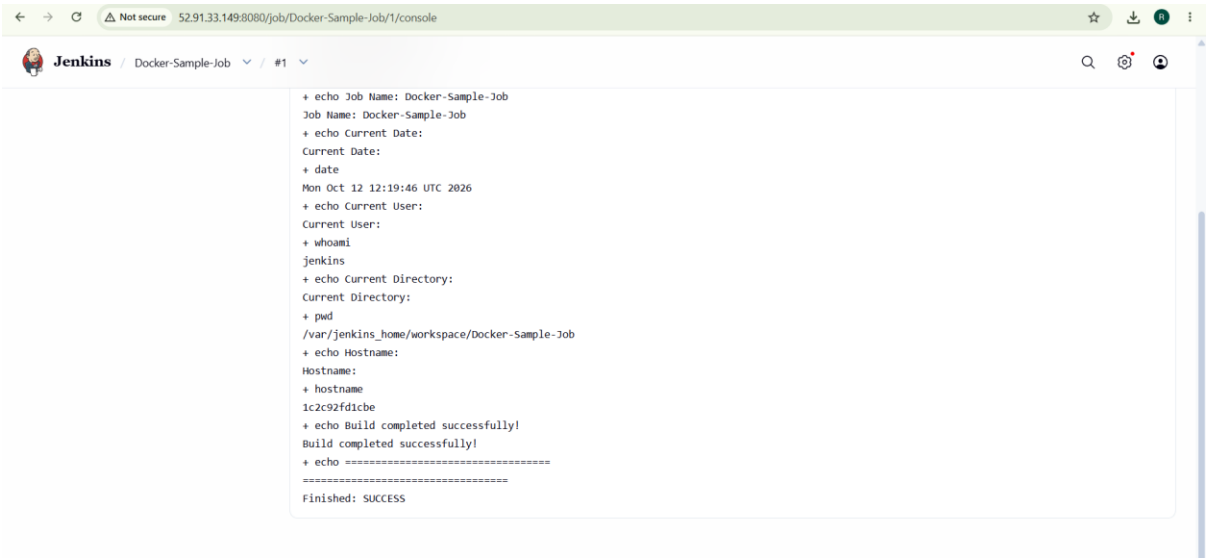

Not secure 52.91.33.149:8080
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** JSON Path API
** Structs
** Pipeline: Step API
** commons-text API
** Token Macro
Build Timeout
** bouncycastle API
** - required dependency
Jenkins 2.580.1


Not secure 52.91.33.149:8080
Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status 0/2
Add description
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
Learn more about distributed builds
REST API
Jenkins 2.580.1


Not secure 52.91.33.149:8080/view/all/newJob
Jenkins / All / New Item
New Item
Enter an item name
Docker-Sample-Job
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
OK


Not secure 52.91.33.149:8080/job/Docker-Sample-Job/configure
Jenkins / Docker-Sample-Job / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Add timestamps to the Console Output
Inspect build log for published build scans
Terminate a build if it's stuck
With Ant
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Execute shell
Command
See the list of available environment variables
pwd
echo "Hostname:"
hostname
echo "Build completed successfully!"
echo "================================="
Advanced
Save
Apply


Not secure 52.91.33.149:8080/job/Docker-Sample-Job/1/console
Jenkins / Docker-Sample-Job / #1
Status
Changes
Console Output
Delete build '#1'
Edit build information
Console
Download
Copy
View as plain text
Started by user Poojith
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Docker-Sample-Job
[Docker-Sample-Job] $ /bin/sh -xe /tmp/jenkins5286902353754845935.sh
+ echo ==========
==========
+ echo Docker Jenkins Sample Job
Docker Jenkins Sample Job
+ echo Build Number: 1
Build Number: 1
+ echo Job Name: Docker-Sample-Job
Job Name: Docker-Sample-Job
+ echo Current Date:
Current Date:
+ date
Mon Oct 12 12:19:46 UTC 2026
+ echo Current User:
Current User:
+ whoami
jenkins
+ echo Current Directory:
Current Directory:
+ pwd
/var/jenkins_home/workspace/Docker-Sample-Job
+ echo Hostname:


Not secure 52.91.33.149:8080/job/Docker-Sample-Job/1/console
Jenkins / Docker-Sample-Job / #1
+ echo Job Name: Docker-Sample-Job
Job Name: Docker-Sample-Job
+ echo Current Date:
Current Date:
+ date
Mon Oct 12 12:19:46 UTC 2026
+ echo Current User:
Current User:
+ whoami
jenkins
+ echo Current Directory:
Current Directory:
+ pwd
/var/jenkins_home/workspace/Docker-Sample-Job
+ echo Hostname:
Hostname:
+ hostname
1c2c92fd1cbe
+ echo Build completed successfully!
Build completed successfully!
+ echo ==========
==========
Finished: SUCCESS